

SOUTHAMPTON, United Kingdom

WMO: 038650

Lat: 50.950N

Lon: 1.357W

Elev: 44

StdP: 14.67

Time Zone: 0.00 (EUW)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	30.6	20.2	15.2	35.2	23.1	17.5	36.4	24.4	48.8	21.1	49.0	4.7	50	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.4	79.2	64.9	76.1	63.3	73.3	62.4	66.9	75.3	65.3	73.0	63.9	70.7	6.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	88.9	69.6	62.4	84.6	68.1	61.1	80.7	67.0	31.4	75.4	30.2	73.1	29.1	70.7	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	15.9	13.8	DB	23.5	84.6	3.0	3.5	21.3	87.2	19.6	89.3	17.9	91.3	15.7	93.8	(4)
(5)			WB	22.3	71.3	3.2	3.7	20.0	73.9	18.2	76.1	16.4	78.2	14.0	80.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	43.1	43.5	46.8	49.2	56.1	59.9	64.4	64.8	59.5	53.5	47.6	43.4
	DBStd	9.20	5.46	5.51	4.36	4.59	5.46	4.16	3.88	4.30	3.95	5.03	5.38	6.04
	HDD50	943	217	184	115	69	10	0	0	0	0	24	107	217
	HDD65	4596	678	601	564	473	281	163	59	56	170	357	523	671
	CDD50	1931	4	3	16	46	197	297	445	459	285	133	35	11
	CDD65	109	0	0	0	0	4	9	40	50	5	1	0	0
	CDH74	515	0	0	0	0	32	52	188	227	15	1	0	0
	CDH80	71	0	0	0	0	1	6	20	44	0	0	0	0
Wind	WSAvg	5.7	6.8	6.6	6.1	6.4	5.6	5.4	4.9	4.7	5.0	5.6	5.5	6.4
Precipitation	PrecAvg	32.1	3.7	2.6	2.0	2.2	1.9	2.1	1.9	2.2	2.3	3.7	4.0	3.5
	PrecMax	44.2	9.3	6.6	5.3	5.9	4.1	5.7	4.6	5.3	5.5	6.6	8.5	7.1
	PrecMin	24.4	0.7	0.2	0.2	0.0	0.3	0.6	0.5	0.2	0.2	1.1	1.0	0.9
	PrecStd	5.2	1.9	1.7	1.1	1.4	1.1	1.4	1.1	1.3	1.3	1.6	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.9	55.5	60.2	67.3	77.8	79.7	83.0	85.4	76.3	68.5	60.4	56.6
		MCWB	50.6	51.3	51.7	54.9	61.0	63.5	66.2	65.4	63.9	60.4	56.5	53.4
	2%	DB	53.4	53.3	57.3	62.8	73.3	74.3	78.9	79.6	71.5	64.6	58.4	55.4
		MCWB	50.6	49.4	50.0	52.4	59.3	61.7	64.5	66.1	61.3	58.3	55.6	53.0
	5%	DB	52.2	52.0	55.1	59.6	69.4	71.0	75.7	75.8	68.9	62.4	57.1	54.1
		MCWB	49.9	48.3	48.8	51.4	58.0	59.6	63.2	64.4	60.1	57.4	54.4	51.7
	10%	DB	51.0	50.8	53.3	56.8	65.7	68.1	72.8	72.9	66.5	60.9	55.5	52.4
		MCWB	48.7	47.4	48.1	49.4	56.3	58.2	62.0	63.1	58.8	56.5	52.8	50.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	52.7	53.5	56.9	63.9	65.7	69.0	69.6	65.4	61.9	57.6	54.7
		MCDB	53.3	54.5	58.3	63.8	73.5	77.6	78.6	79.4	72.4	66.0	58.7	55.7
	2%	WB	51.3	50.8	51.9	54.0	61.0	62.8	66.4	67.4	63.6	60.1	56.5	53.5
		MCDB	52.6	52.4	55.2	60.1	70.2	71.6	75.4	75.7	69.3	62.9	57.9	54.9
	5%	WB	50.3	49.5	50.8	52.3	59.1	61.0	64.7	66.0	62.1	58.9	54.9	52.1
		MCDB	52.0	51.2	53.6	57.3	67.6	68.4	72.5	73.3	66.6	61.4	56.6	53.8
	10%	WB	49.0	48.1	49.6	50.9	57.1	59.5	63.3	64.6	60.7	57.3	53.4	50.4
		MCDB	50.8	50.2	52.2	55.2	64.2	65.9	70.4	71.2	64.8	60.4	55.2	52.3
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.2	10.7	12.7	14.4	13.4	14.3	14.4	13.0	11.3	9.7	8.5
		MCDBR	8.3	9.3	13.0	16.4	20.7	18.6	20.0	19.1	16.3	12.0	8.4	7.7
		MCWBR	8.0	7.8	8.0	9.0	10.7	9.3	9.3	9.3	9.1	8.1	7.1	7.8
	5% WB	MCDBR	8.6	8.9	10.4	14.0	18.3	16.5	16.2	16.3	14.3	10.5	8.1	7.8
		MCWBR	8.5	8.5	7.4	8.7	10.4	9.3	8.9	9.3	9.4	8.2	7.6	8.3
Clear-Sky Solar Irradiance	taub	0.332	0.347	0.384	0.405	0.408	0.404	0.397	0.390	0.376	0.361	0.333	0.322	
	taud	2.360	2.367	2.260	2.226	2.260	2.300	2.326	2.366	2.394	2.414	2.406	2.346	
	Ebn at Noon	216	246	256	264	269	271	271	267	257	238	214	199	
	Edh at Noon	21	26	35	40	41	39	38	35	30	25	20	18	
All-Sky Solar Radiation	RadAvg	262	485	858	1353	1633	1807	1715	1433	1065	606	318	206	
	RadStd	20	59	111	134	99	172	143	110	82	50	36	24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page